Solar Si appendix

Sun · Si · downloadable appendix

Document	Solar Si appendix
Subject	Solar photospheric abundance appendix · Si
Author	Ryan Schmitt
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Solar silicon: 7.517 ± 0.036

Asplund, Amarsi & Grevesse 2021: 7.51 ± 0.03 · this measurement +0.007 dex

Measured the way Asplund, Amarsi & Grevesse (2021) measure it: the Asplund et al. 2021 line set (Amarsi & Asplund 2017: nine Si I and one Si II line), 10 lines measured on one spectrum, each part weighted by its uncertainty.

Si I: 7.486 ± 0.037 · Kitt Peak (Kurucz 2005) · VIS · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)

Si II: 7.528 ± 0.152 · Kitt Peak (Kurucz 2005) · VIS · Si II · Asplund et al. 2021 lines · 1D LTE

Si I: 7.563 ± 0.045 · Kitt Peak (Kurucz 2005) · red-optical · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)

Highlighted Products

VIS

7.489 ± 0.036

Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

red-optical

7.563 ± 0.045

Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · Kitt Peak (Kurucz 2005) · Reference Grade

NIR

7.522 ± 0.040

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

H

7.325 ± 0.046

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · CRIRES+ H · Reference Grade

Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	35 products -- Si I · Asplund et al. 2021 lines; Si I · Deshmukh et al. 2022 lines; Si I · Elgueta et al. 2026 lines; Si II · Asplund et al. 2021 lines

Grade	What it means	This element
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	21 products -- Si I · Codex-graded lines; Si II · Codex-graded lines
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

VIS — HARPS
A(Si) dex; axis 7.15–7.84

Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.535 ±0.151
Si II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		7.511 ±0.043
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.514 ±0.056
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.487 ±0.057
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.514 ±0.056
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.487 ±0.057

VIS — Kitt Peak (Kurucz 2005)
A(Si) dex; axis 7.15–7.84

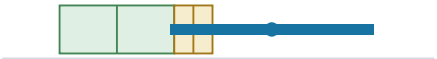
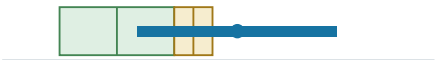
Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.528 ±0.152
Si II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.491 ±0.315
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.513 ±0.036
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.486 ±0.037
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.513 ±0.036
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.486 ±0.037

VIS — Kitt Peak (1984, molecfit)
A(Si) dex; axis 7.15–7.84

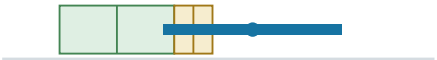
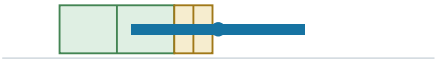
Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.518 ±0.117
Si II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.488 ±0.287
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.516 ±0.036
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.489 ±0.036
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.516 ±0.036
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.489 ±0.036

VIS axis: 7.15 · 7.32 · 7.49 · 7.66 · 7.84 A(Si) dex

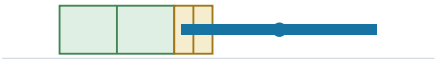
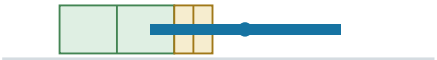
red-optical — IAG FTS
A(Si) dex; axis 7.45–7.68

Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.591 ±0.053
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.573 ±0.052

red-optical — Kitt Peak (Kurucz 2005)
A(Si) dex; axis 7.45–7.68

Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.581 ±0.047
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.563 ±0.045

red-optical — Kitt Peak (1984, molecfit)
A(Si) dex; axis 7.45–7.68

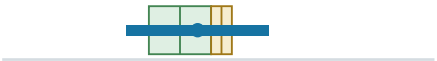
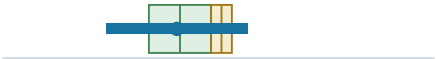
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.595 ±0.051
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.577 ±0.050

red-optical axis: 7.45 · 7.51 · 7.56 · 7.62 · 7.68 A(Si) dex

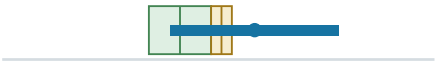
NIR — CRIRES+ J
A(Si) dex; axis 7.34–7.76

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 2 lines		7.513 ±0.032

NIR — CRIRES+ J
A(Si) dex; axis 7.34–7.76

Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.487 ±0.033
Si I · Codex-graded lines · 1D LTE Codex Grade · 27 lines		7.527 ±0.069
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 27 lines		7.507 ±0.068

NIR — CRIRES+ Y
A(Si) dex; axis 7.34–7.76

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 1 line		7.468 ±0.035
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.428 ±0.035
Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 2 lines		7.457 ±0.060
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.430 ±0.059
Si I · Codex-graded lines · 1D LTE Codex Grade · 4 lines		7.582 ±0.081
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 4 lines		7.560 ±0.086

NIR — CRIRES+ Y (full arm)
A(Si) dex; axis 7.34–7.76

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 1 line		7.468 ±0.035
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.428 ±0.035
Si I · Codex-graded lines · 1D LTE Codex Grade · 8 lines		7.551 ±0.065
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 8 lines		7.523 ±0.067

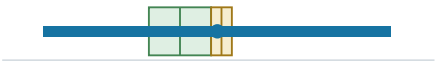
NIR — IAG FTS
A(Si) dex; axis 7.34–7.76

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 1 line		7.554 ±0.129
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.542 ±0.129

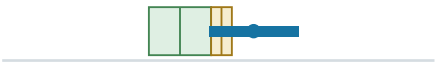
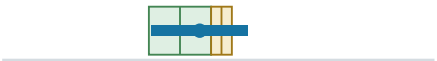
NIR — IAG FTS
A(Si) dex; axis 7.34–7.76

Si I · Codex-graded lines · 1D LTE Codex Grade · 13 lines		7.528 ±0.093
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 9 lines		7.497 ±0.128

NIR — Kitt Peak (Kurucz 2005)
A(Si) dex; axis 7.34–7.76

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 1 line		7.558 ±0.167
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.546 ±0.167
Si I · Codex-graded lines · 1D LTE Codex Grade · 13 lines		7.558 ±0.054
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 9 lines		7.547 ±0.061

NIR — Kitt Peak (1984, molecfit)
A(Si) dex; axis 7.34–7.76

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 3 lines		7.581 ±0.043
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.552 ±0.047
Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 15 lines		7.536 ±0.041
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 15 lines		7.522 ±0.040
Si I · Codex-graded lines · 1D LTE Codex Grade · 46 lines		7.536 ±0.045
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 43 lines		7.529 ±0.046

NIR axis: 7.34 · 7.44 · 7.55 · 7.65 · 7.76 A(Si) dex

H — CRIRES+ H
A(Si) dex; axis 7.25–7.59

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 3 lines		7.363 ±0.049
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.325 ±0.046

H — CRIRES+ H (full arm, telluric-corrected)
A(Si) dex; axis 7.25–7.59

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 3 lines		7.418 ±0.062
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.380 ±0.056

H axis: 7.25 · 7.33 · 7.42 · 7.50 · 7.59 A(Si) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every silicon value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the silicon abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Si)	Largest uncertainty terms (dex)
VIS	Si I · Codex-graded lines; Si II · Codex-graded lines	HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2–7	7.486–7.489 (1D-NLTE)	line strengths (gf) 0.05, continuum placement 0.03
VIS	Si I · Asplund et al. 2021 lines; Si II · Asplund et al. 2021 lines	HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	1–7	7.486–7.489 (1D-NLTE)	line strengths (gf) 0.05, continuum placement 0.03
red-optical	Si I · Asplund et al. 2021 lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2	7.563–7.577 (1D-NLTE)	line strengths (gf) 0.11, fit / line-to-line scatter 0.03
NIR	Si I · Codex-graded lines	CRIRES+ J, CRIRES+ Y, CRIRES+ Y (full arm), IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	4–46	7.497–7.560 (1D-NLTE)	telluric residual 0.55, line strengths (gf) 0.04

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Si)	Largest uncertainty terms (dex)
NIR	Si I · Deshmukh et al. 2022 lines	CRIRES+ J, CRIRES+ Y, CRIRES+ Y (full arm), Kitt Peak (1984, molecfit)	1D-LTE, 1D-NLTE	1–3	7.428–7.552 (1D-NLTE)	line strengths (gf) 0.06, fit / line-to-line scatter 0.04
NIR	Si I · Elgueta et al. 2026 lines	CRIRES+ Y, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	1–15	7.430–7.546 (1D-NLTE)	line strengths (gf) 0.04, fit window 0.03
H	Si I · Elgueta et al. 2026 lines	CRIRES+ H, CRIRES+ H (full arm, telluric-corrected)	1D-LTE, 1D-NLTE	3	7.325–7.380 (1D-NLTE)	continuum placement 0.06, line strengths (gf) 0.05

Excluded lines

These lines are left out of the Sun's measurement. They stay in the line lists: a cooler or more evolved star may measure them well. Blended lines are not excluded -- the spectrum synthesis models the blend. 59 lines.

Abundance outlier: disagrees with the other lines well beyond its error, and no solar analysis in the literature uses it.

Not measurable: the line falls in a data gap or a saturated telluric band.

Saturated: on the flat part of the curve of growth, so the line barely responds to the abundance (reduced width log W/λ above -4.9).

Species	Wavelength (Å, air)	Why it is excluded
Si II	3203.872	saturated (log W/λ = -4.18)
Si II	3210.028	saturated (log W/λ = -3.98)
Si II	3339.818	saturated (log W/λ = -4.04)
Si II	3856.017	saturated (log W/λ = -3.90)
Si II	3862.595	saturated (log W/λ = -4.15)
Si I	3905.523	saturated (log W/λ = -3.85)
Si I	4102.936	saturated (log W/λ = -4.10)
Si I	5948.541	saturated (log W/λ = -4.79)
Si I	7918.383	saturated (log W/λ = -4.88)
Si I	7932.348	saturated (log W/λ = -4.81)
Si I	7944.001	saturated (log W/λ = -4.75)
Si I	8543.151	saturated (log W/λ = -4.23)
Si I	8648.465	saturated (log W/λ = -4.72)
Si I	8661.177	saturated (log W/λ = -4.27)
Si I	8728.010	saturated (log W/λ = -4.89)
Si I	8742.446	saturated (log W/λ = -4.90)
Si I	8752.007	saturated (log W/λ = -4.79)
Si I	8790.389	saturated (log W/λ = -4.82)
Si I	9021.590	saturated (log W/λ = -4.88)
Si I	9413.500	saturated (log W/λ = -4.56)
Si I	9887.047	abundance outlier
Si I	10313.197	abundance outlier

Species	Wavelength (Å, air)	Why it is excluded
Si I	10371.263	saturated (log W/λ = -4.73)
Si I	10585.141	saturated (log W/λ = -4.46)
Si I	10603.425	saturated (log W/λ = -4.57)
Si I	10627.648	saturated (log W/λ = -4.85)
Si I	10660.973	saturated (log W/λ = -4.53)
Si I	10689.716	saturated (log W/λ = -4.70)
Si I	10694.251	saturated (log W/λ = -4.66)
Si I	10727.406	saturated (log W/λ = -4.62)
Si I	10749.378	saturated (log W/λ = -4.54)
Si I	10786.849	saturated (log W/λ = -4.59)
Si I	10827.088	saturated (log W/λ = -4.40)
Si I	10843.858	saturated (log W/λ = -4.73)
Si I	10868.790	saturated (log W/λ = -4.49)
Si I	10869.536	saturated (log W/λ = -4.34)
Si I	10885.333	saturated (log W/λ = -4.84)
Si I	10979.308	saturated (log W/λ = -4.62)
Si I	10982.058	saturated (log W/λ = -4.90)
Si I	11017.966	saturated (log W/λ = -4.68)
Si I	11187.588	saturated (log W/λ = -4.89)
Si I	11289.829	saturated (log W/λ = -4.86)
Si I	11611.060	saturated (log W/λ = -4.60)
Si I	11984.198	saturated (log W/λ = -4.46)
Si I	11991.568	saturated (log W/λ = -4.51)
Si I	12031.504	saturated (log W/λ = -4.38)
Si I	12103.534	saturated (log W/λ = -4.59)
Si I	12189.241	abundance outlier
Si I	12270.692	saturated (log W/λ = -4.62)
Si I	15077.332	saturated (log W/λ = -4.67)
Si I	15557.778	saturated (log W/λ = -4.88)
Si I	15888.409	saturated (log W/λ = -4.39)
Si I	15960.063	saturated (log W/λ = -4.53)
Si I	16060.009	saturated (log W/λ = -4.64)
Si I	16163.691	saturated (log W/λ = -4.78)
Si I	16380.176	saturated (log W/λ = -4.72)
Si I	16381.534	saturated (log W/λ = -4.54)
Si I	16434.927	not measurable
Si I	17466.879	saturated (log W/λ = -4.45)

Solar silicon context

Silicon is traced by Si I lines from the blue to the H band and by the single Si II 6371 Å line. The optical lines are Asplund et al. 2021's set (Amarsi & Asplund 2017) with the gf the authors used; the infrared lines are the published sets of Deshmukh et al. 2022 and Elgueta et al. 2026 on Pehlivan Rhodin et al. 2024 oscillator strengths. Each set, spectrum and treatment remains a separate product.

Published line sets · gf uncertainty model · Methodology

Download the evidence

Download PDF — Solar Si appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit f5c6b3d35f016dd66b74d3a098459bd648f9677f
Feed Si.json v1.279 · updated 2026-10-29T03:00:33Z
Generated 2026-10-28T21:25:21-06:00 (source commit timestamp)
Feed SHA-256 c410d23996df5c4ea8521d52052c547837348d49f209b44819cc281d1147cdb9
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar Si comparison literature; not an adopted Codex abundance. Sources: [1], [2].

Solar abundance and indicator analyses. Published line set SI_AGSS21 (the line selection). Sources: [3].

Atomic data and gf authorities. Si II oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [3], [4], [5].

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [6].

Solar abundance and indicator analyses. Published line set SI_ELGUETA2026 (the line selection). Sources: [8].

Atomic data and gf authorities. Si I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [9].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Si_Amarsi2020_PySME.csv). Sources: [10].

Solar abundance and indicator analyses. Published line set SI_DESHMUKH2022 (the line selection). Sources: [11].

Product-to-source index

VIS · HARPS · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-18313ebf6e74

VIS · HARPS · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6] Si-products-a600a1f3ca00

VIS · Kitt Peak (Kurucz 2005) · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-7b250bcdf59f

VIS · Kitt Peak (Kurucz 2005) · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6] Si-products-8d7dbfa460a5

VIS · Kitt Peak (1984, molecfit) · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-7f5a70dae4b6

VIS · Kitt Peak (1984, molecfit) · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6] Si-products-d48b7d00757a

H · CRIRES+ H · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-ae368e68c20f

H · CRIRES+ H · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-dce72b125a33

H · CRIRES+ H (full arm, telluric-corrected) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-35e19a8b409b

H · CRIRES+ H (full arm, telluric-corrected) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-f15f24015969

NIR · CRIRES+ J · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-f4f1f4acdbbd

NIR · CRIRES+ J · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10] Si-products-2f8d4150cd36

NIR · CRIRES+ J · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-32aa007f7e8f

NIR · CRIRES+ J · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-12ecd615a57c

NIR · CRIRES+ Y · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-2e6494caa930

NIR · CRIRES+ Y · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10] Si-products-7279b403d536

NIR · CRIRES+ Y · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-7bf51922dcd8

NIR · CRIRES+ Y · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-9f171f68364e

NIR · CRIRES+ Y · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-d9d944a5174a

NIR · CRIRES+ Y · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-9d2cc2180104

NIR · CRIRES+ Y (full arm) · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-4118112da051

NIR · CRIRES+ Y (full arm) · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10] Si-products-62daec519fc5

NIR · CRIRES+ Y (full arm) · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-ae2252e2595e

NIR · CRIRES+ Y (full arm) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-39fe1f0c36ba

NIR · IAG FTS · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-aecbf88dfbea

NIR · IAG FTS · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-acb5d5cc712d

NIR · IAG FTS · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-db04798aba7f

NIR · IAG FTS · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-ca0fde457ba5

NIR · Kitt Peak (Kurucz 2005) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-8179b4d2e5e5

NIR · Kitt Peak (Kurucz 2005) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-4ff484ea11a9

NIR · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-17e97411dd39

NIR · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-d57ff6e80885

NIR · Kitt Peak (1984, molecfit) · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-a2f3e8d83fe2

NIR · Kitt Peak (1984, molecfit) · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10] Si-products-a495ceeb903a

NIR · Kitt Peak (1984, molecfit) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-d33e7d22bb91

NIR · Kitt Peak (1984, molecfit) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-ef61cd529b70

NIR · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-73a50431758c

NIR · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-d7488f68293c

VIS · HARPS · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-bf1d419e9653

VIS · HARPS · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-f7ba31ede52a

VIS · HARPS · Si I · Codex-graded lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-1f64e95aec54

VIS · HARPS · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-e4e2c2a05b2c

VIS · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-d5852ff8d906

VIS · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-45bffeb3d297

VIS · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-c620313cd86a

VIS · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-86de3ad0a76e

VIS · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-9dbedbf77e52

VIS · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-8f7445cbe3bd

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VIS · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-a7d70038b8ec

red-optical · IAG FTS · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-412125424711

red-optical · IAG FTS · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-8a4cd1b740bf

red-optical · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-60f8513260fd

red-optical · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-309c3f636467

red-optical · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-28bfa81eb8b3

red-optical · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-5d0cd9de9939

NIR · CRIRES+ Y (full arm) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-quarantine-7ebb8d8816da

NIR · CRIRES+ Y (full arm) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-quarantine-c8be7c1b1687

NIR · CRIRES+ J · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-quarantine-6171b736d381

NIR · CRIRES+ J · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-quarantine-b905b0754458

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. A&A 653, A141 DOI: 10.1051/0004-6361/202140445. source record

[2] — Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites. Space Science Reviews 221 DOI: 10.1007/s11214-025-01146-w. source record

[3] — Amarsi, A. M.; Asplund, M. (2017). The solar silicon abundance based on 3D non-LTE calculations. MNRAS 464, 264-273 DOI: 10.1093/mnras/stw2445. source record

[8] — Elgueta, S.; Jofre, P.; Aguilera-Gomez, C.; Slumstrup, D.; Rojas Arriagada, A.; Heiter, U.; Casamiquela, L.; Zoccali, M.; Worley, C.; Soubiran, C. (2026). Gaia FGK benchmark stars: Selecting infrared lines for abundance determination. A&A 710, A111 DOI: 10.1051/0004-6361/202659148. source record

[11] — Deshmukh, S. A.; Ludwig, H.-G.; Kucinskas, A.; et al. (2022). The solar photospheric silicon abundance according to CO5BOLD: Investigating line broadening, magnetic fields, and model effects. A&A 668, A48 DOI: 10.1051/0004-6361/202142072. source record

Atomic data and gf authorities

[4] — Garz, T. (1973). Absolute oscillator strengths of Si I lines between 2500 A and 8000 A. A&A 26, 471-477 source record

[5] — O'Brian, T. R.; Wickliffe, M. E.; Lawler, J. E.; Whaling, W.; Brault, J. W. (1991). Lifetimes, transition probabilities, and level energies in Fe I. JOSA B 8, 1185 DOI: 10.1364/JOSAB.8.001185. source record

[7] — Kelleher, D. E.; Podobedova, L. I. (2008). Atomic Transition Probabilities of Silicon. A Critical Compilation. J. Phys. Chem. Ref. Data 37, 1285-1501 DOI: 10.1063/1.2734566. source record

[9] — Pehlivan Rhodin, A.; Hartman, H.; Nilsson, H.; Joensson, P. (2024). Accurate and experimentally validated transition data for Si I and Si II. A&A 682, A184 DOI: 10.1051/0004-6361/202245686. source record

Model and grid sources

[6] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. source record

[10] — Amarsi, A. M.; Lind, K.; Osorio, Y.; Nordlander, T.; Bergemann, M.; et al. (2020). The GALAH Survey: non-LTE departure coefficients for large spectroscopic surveys. A&A 642, A62 DOI: 10.1051/0004-6361/202038650. source record